🧠 BUSINESS CONTEXT

You are a **Data Engineer** supporting an **online shopping platform**.

The business team wants a **clean, optimized dataset** to analyze:

- Order revenue
- City-wise performance
- Product demand

Your task is to **clean, analyze, optimize, and store order data using PySpark**.

📂 DATASET PROVIDED

DATASET — ONLINE ORDERS

```
orders_data = [
    ("O001","Delhi ","Laptop","45000","2024-01-05","Completed"),
    ("O002","Mumbai","Mobile ","32000","05/01/2024","Completed"),
    ("O003","Bangalore","Tablet","30000","2024/01/06","Completed"),
    ("O004","Delhi","Laptop","","2024-01-07","Cancelled"),
    ("O005","Mumbai","Mobile","invalid","2024-01-08","Completed"),
    ("O006","Chennai","Tablet",None,"2024-01-08","Completed"),
    ("O007","Delhi","Laptop","47000","09-01-2024","Completed"),
    ("O008","Bangalore","Mobile","28000","2024-01-09","Completed"),
    ("O009","Mumbai","Laptop","55000","2024-01-10","Completed"),
    ("O009","Mumbai","Laptop","55000","2024-01-10","Completed")
]
```

Columns

```
order_id
city
product
amount
order_date
status
```

🎯 PROJECT OBJECTIVES

PHASE 1 — DATA INGESTION & SCHEMA

Topics:
StructType, StructField, data types

Tasks

1. Define an explicit schema
2. Create a DataFrame using the schema
3. Print schema and validate data types

PHASE 2 — DATA CLEANING

Topics:
Column ops, filter, withColumn, replace

Tasks

4. Trim all string columns
5. Standardize `city` and `product` values
6. Convert `amount` to IntegerType
7. Handle invalid and null `amount` values
8. Remove duplicate orders
9. Keep only `Completed` orders

PHASE 3 — BASIC ANALYTICS

Topics:
GroupBy, aggregations

Tasks

10. Total revenue per city

11. Total revenue per product
12. Average order value per city

PHASE 4 — WINDOW FUNCTION

Topics:
Window, rank

Tasks

13. Rank cities by total revenue
14. Identify top-performing city

(One window only — no complex nesting)

PHASE 5 — PERFORMANCE AWARENESS

Topics:
Cache, explain, partitions

Tasks

15. Cache the cleaned DataFrame
16. Run two aggregations and observe behavior
17. Use `explain(True)` to inspect the plan
